

Safety Outcome: Time Spent Near Singularities

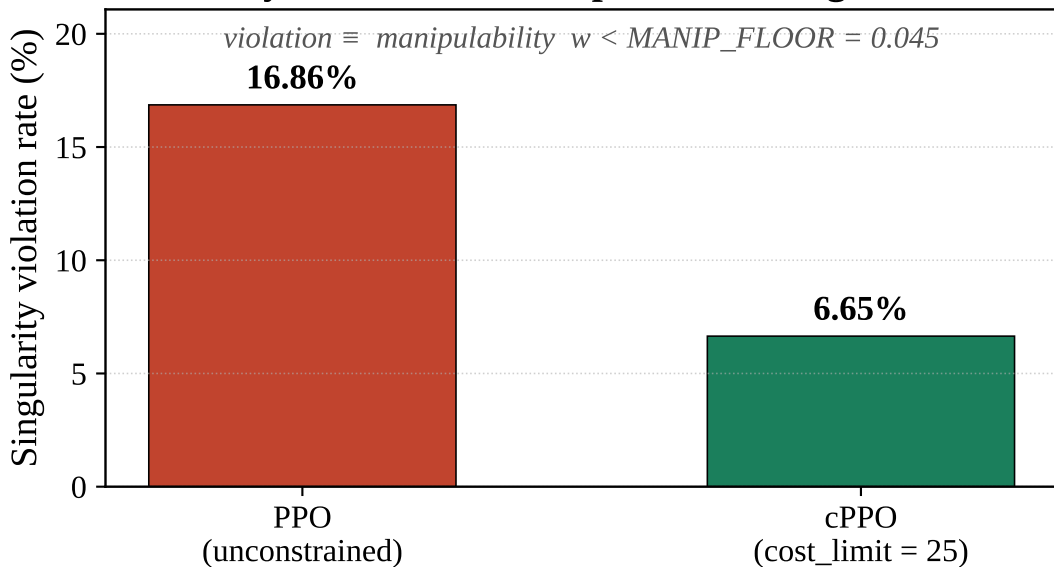


Figure 4. Final singularity violation rate (a violation is $w < MANIP_FLOOR = 0.045$). cPPO more than halves the time spent near kinematic singularities versus unconstrained PPO (6.65% vs. 16.86%).